Leo Yao

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Education

Carnegie Mellon University August 2024 – May 2028
B.S. in Computer Science and Statistics & Machine Learning Pittsburgh, PA

- **QPA:** 3.65/4.00
- **Relevant Coursework:** Computer Systems, Parallel & Sequential Data Structures & Algorithms, Computer Security, Generative AI (Graduate), Machine Learning, Statistical Computing, Functional Programming, Theoretical Computer Science
- **Awards:** Dean's List, High Honors

Experience

Software Developer Intern May 2026 - August 2026
IBM San Jose, CA

- Built an AI-agent observability connector for watsonx.governance, integrating three vendor APIs (Arrow Flight/gRPC, GraphQL, REST) to normalize per-agent cost, latency, error-rate, and LLM evaluator scores.
- Instrumented multi-agent LangGraph systems on two observability platforms (Arize AX, Phoenix) with OpenTelemetry/OpenInference, emitting all 8 span kinds plus PII/guardrail spans.
- Cut peak worker memory 6.4× by chunking unbounded, customer-controlled span exports, and reduced live-path API calls 22→1 (∼95%) by redesigning around a ∼1,000 query-complexity cap.
- Wrote 373 tests at ∼2:1 test-to-source, including a differential-oracle suite that recomputes every metric independently over 1,500 randomized inputs and 16,000 percentile comparisons.

Teaching Assistant January 2026 - May 2026
Carnegie Mellon University Pittsburgh, PA

- Led weekly 1.5-hour recitations and 3 hours of office hours for 20+ students in 15-150 Principles of Functional Programming, CMU's required course on SML, recursion, and proofs of program correctness.
- Graded 300+ homework and exam submissions weekly in SML, giving technical feedback on correctness, efficiency, and style across recursive, higher-order, and parallel implementations.

Research Assistant September 2025 - May 2026
Carnegie Mellon University Pittsburgh, PA

- Owned the training and evaluation harness for quantization-aware, FPGA-bound GNNs at CERN CMS: resumable checkpointing, CLI-driven hyperparameter sweeps, and evaluation against the PUPPI baseline.
- Adapted lightweight PyTorch transformer models for use in particle clustering on CERN HGCAL data, creating a custom data pipeline with NumPy and Awkward to properly load in data for training.

Projects

PharmaSentinel | *Python, PostgreSQL, Supabase, OpenAI API* February 2026

- Engineered a multi-agent backend system in Python, orchestrating external REST APIs (OpenAI, openFDA) to automate real-time supply chain monitoring and substitute identification.
- Architected event-driven PostgreSQL databases utilizing Supabase Realtime to detect inventory state changes, automatically triggering agent pipelines and routing resolutions to downstream clients.

Computer Systems | *C, POSIX threads* May 2025 – August 2025

- **Dynamic Memory Allocator**: Developed a dynamic memory allocator in C for Linux. Achieved 74.3% memory utilization and a throughput of 15885 kilo-operations per second, ranking at the top of the class.
- **Tiny Linux Shell**: Engineered a Linux shell featuring foreground/background job control, I/O redirection, and robust signal handling to manage concurrent process execution.
- **Shark File System**: Built a thread-safe concurrent file system with file renaming and descriptor position management, using mutex synchronization for multithreaded correctness.

Skills

Programming Languages: Python, C, C++, SQL, SML, x86-64 Assembly
Backend & APIs: FastAPI, asyncio, Redis, GraphQL, REST, PostgreSQL, Supabase, Docker, Linux
AI & Observability: LangGraph, OpenTelemetry, OpenInference, Arize AX/Phoenix, MCP, OpenAI API
Data & Testing: pandas, NumPy, PyTorch, Apache Arrow Flight, pytest, Git, GDB, Valgrind